Change Report & Implementation Summary

6-DOF Robotic Arm Pick-and-Place and Color Sorting

Repository: https://github.com/saumyadebug/robotic-arm-pick-and-place

Report source: 24BRS1075_24BRS1065_24BRS1126_Poster_A3 (7).pdf

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1. Purpose of This Work

The project report (poster) defines the full target system (6-DOF, 4 color bins). For this iteration the priority is narrower: make the arm move reliably across all joints, make the gripper work properly, prevent mechanical collision/collapse, and complete a stable pick-and-place cycle. Yellow sorting and the extra (6th) DOF are deferred until base motion is stable. This work updates firmware with safe motion and per-joint limits, documentation, and test records, and delivers an implementation plan plus this change report.

2. What Was Changed (and Why)

Firmware: pick_and_place.ino now moves joints sequentially with per-joint pulse clamping to reduce simultaneous load and collision risk. integrated_sorting.ino uses clamped pulses, debounced ultrasonic triggering, and standardized PCA9685 init; R/G/B sorting retained (Yellow deferred). color_sensor_test.ino reverted to R/G/B. Hardware docs and tests reverted to the stable 4-DOF / R/G/B baseline. README status set to In Progress (stabilization).

3. What Was Added

- Per-joint safe pulse limits and clampPulse() in firmware.
- Sequential joint motion (moveArmSafe) to reduce load/collision.
- Ultrasonic debounce (3 consecutive readings) in STATE_IDLE.
- Filter-settling delay (10 ms) between color filter switches.
- Hardware bring-up checklist in IMPLEMENTATION_PLAN.md.
- IMPLEMENTATION_PLAN.md (root and docs/).
- This change-report PDF.

4. What Was Removed (reverted from prior iteration)

- Wrist roll (6th DOF) channel support.
- Yellow color detection and 4th bin.
- Corresponding test rows (channel 5, Yellow cycle).

5. Per-File Change Details

code/pick_and_place/pick_and_place.ino
- Sequential motion, clamped pulses, standardized init.

code/integrated_sorting/integrated_sorting.ino
- Clamped pulses, debounce, standardized init; R/G/B only.

code/color_sensor_test/color_sensor_test.ino
- Reverted to R/G/B classification.

code/end_effector_test/end_effector_test.ino
- Standardized PCA9685 oscillator frequency.

hardware/components.md
- Voltage regulator specified as XL4016.

hardware/power_supply.md
- Buck converter specified as XL4016.

docs/sensor_integration.md
- Gripper-mounted sensor note.

tests/servo_test_results.md
- 4-DOF baseline restored.

tests/pick_and_place_test_results.md
- R/G/B baseline restored.

README.md
- Status -> In Progress (stabilization).

IMPLEMENTATION_PLAN.md
- New stabilization-focused plan.

docs/implementation_plan.md
- Copy of implementation plan.

6. Implementation Plan Summary

Phase 1 · Firmware Core: 6-DOF + 4-color FSM (completed).

Phase 2 · Hardware Documentation Alignment (completed).

Phase 3 · Test Records & Status (completed).

Phase 4 · Planning & Reporting Artifacts (completed; optional CI/tests remain).

Phase 5 · Future Improvements (backlog): interpolation, IK, conveyor, benchmarks.

7. Risks & Mitigations

- No Arduino CLI available: changes kept minimal and consistent; manual review performed.
- Yellow misclassification under ambient light: shielding & white-balance planned (Phase 5).
- 6th servo mechanical interference: calibrate via joint_calibration before full runs.
- Base servo range (150-600) for 4 bins: targets spaced 200/375/550/590.

8. Acceptance Criteria

- All sketches consistent (standardized init, channel defs).
- FSM supports 4 colors and 6 joints.
- Documentation matches report (6 MG995, XL4016, gripper-mounted sensor, 4 bins).
- Test records include 6 channels and Yellow.
- Implementation plan and change-report PDF delivered.

9. Deliverables

- Updated repository (firmware + docs + tests).
- IMPLEMENTATION_PLAN.md.
- docs/change_report.pdf (this document).
- Git commit and push to origin/main.